

Adaptive - Quality Assessment Generative Adversarial Network (AQA - GAN)

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Introduction

Images in Deep Learning

IMAGE REPRESENTATION

An image is a 3D tensor $H \times W \times C$

- H = Height (pixels)
- W = Width (pixels)
- C = Channels (3 for RGB, 1 for grayscale)

Each pixel value $\in [0, 255]$ (uint8)

Normalized to $[0, 1]$ or $[-1, +1]$ for DL training

HOW DL PROCESSES IMAGES

Convolutional Neural Networks (CNNs) apply learnable filters to extract features from raw pixel values.

- **Low-level features** — edges, textures, gradients
- **Mid-level** — shapes, blobs, regions
- **High-level** — objects, semantics, scene understanding

Filters are learned automatically from data via backpropagation.

Motivation & Challenges with Low-Light Images in Deep Learning



Extreme Darkness

Pixel values near 0 — features invisible, gradients vanish during training



Noise Amplification

Sensors introduce noise in dark conditions — CNNs learn noise patterns instead of content



Color Distortion

Artificial light sources (neon, sodium) cause unnatural color casts across the image



Detail Loss

Fine textures and object boundaries are lost in shadows models produce blurry outputs



Uneven Illumination

Parts of the same image may be very dark while others are overexposed

Literature Review

Phase 1

Classical (1970s–2000s)

- Land & McCann (1971) — Retinex theory: separates reflectance from illumination
- Tomasi & Manduchi (1998) — Bilateral filter: edge-preserving spatial smoothing
- Zuiderveld (1994) — CLAHE: contrast limited adaptive histogram equalization
- Donoho & Johnstone (1994) — Wavelet shrinkage for signal denoising

Phase 2

- Elad & Aharon (2006) — K-SVD dictionary learning for sparse representation
- Dabov et al. (2007) — BM3D: collaborative block-matching denoising
- Freeman et al. (2002) — Learning low-level vision with Markov Random Fields
- Breiman (2001) — Random Forests applied to image patch regression

Phase 3

Deep CNNs (2014–2016)

- Dong et al. (2014) — SRCNN: first end-to-end CNN for super-resolution
- Ronneberger et al. (2015) — U-Net: encoder-decoder with skip connections
- Zhang et al. (2017) — DnCNN: residual learning surpasses BM3D
- Kim et al. (2016) — VDSR: very deep networks with residual learning

Phase 4

GAN Era (2017–Present)

- Goodfellow et al. (2014) — Generative Adversarial Networks (GANs)
- Ledig et al. (2017) — SRGAN: perceptual loss + adversarial for photo-realistic SR
- Loh & Chan (2019) — ExDark: first exclusively dark image dataset
- Wang et al. (2021) — EnlightenGAN: unsupervised low-light enhancement

Problem Definition

CURRENT FORMULATION

(Problem with Existing Models)

Current enhancement models apply:

$$\hat{I} = F(I_l; \theta)$$

where:

- I_l = low-light image
- F = enhancement model

Problem:

Same transformation is applied to all images **regardless of quality.**

PROPOSED ADAPTIVE FORMULATION

(Quality-Aware Enhancement)

Proposed adaptive formulation

$$\hat{I} = F(I_l, q; \theta)$$

where:

- I_l = low-light image
- q = image quality level
- F = enhancement model

Quality-aware enhancement adapts according to **degradation severity.**

QUALITY LEVELS

$q = 1$

Low Quality → Aggressive enhancement
(Severely degraded images)

$q = 2$

Medium Quality → Moderate enhancement
(Moderately degraded images)

$q = 3$

High Quality → Subtle enhancement
(Slightly degraded / good quality images)

Objective (s) of the Project

Main Objective

Develop a Quality-Aware GAN framework for adaptive nighttime image enhancement.

Specific Objectives

- Design a Quality Assessment Module (QAM)
- Categorize images using NIQE scores
- Build quality-conditioned generator and discriminator
- Preserve textures and colors
- Reduce over-enhancement artifacts
- Improve perceptual quality metrics:
 - PSNR
 - SSIM

Dataset(s) Used

Dataset Details

1. LOL-v2 Dataset

Contains paired:

- Low-light images
- Ground-truth normal-light images

2. Low-Light Image Enhancement Dataset

Dataset Source:

Kaggle Low-Light Dataset

Preprocessing

- Resize to 256×256
- Random crop
- Horizontal flipping
- Normalization to $[-1, 1]$

Sample 1

Low-light
Image

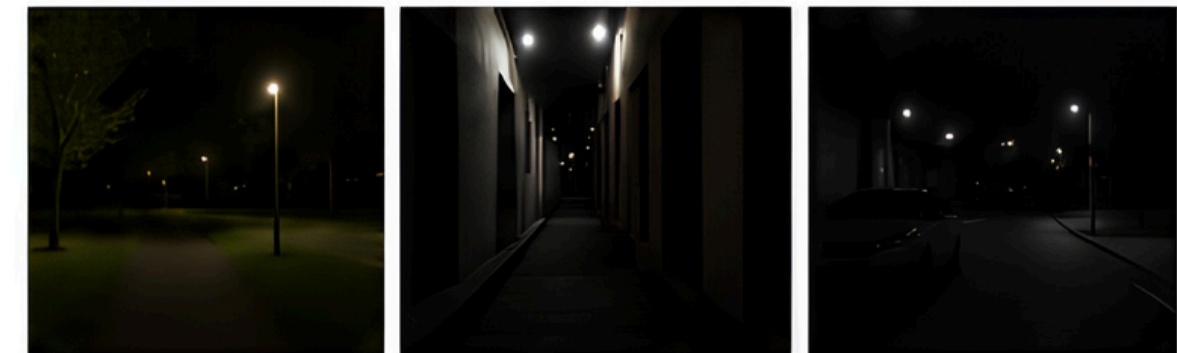


Ground-truth
Normal-light
Image

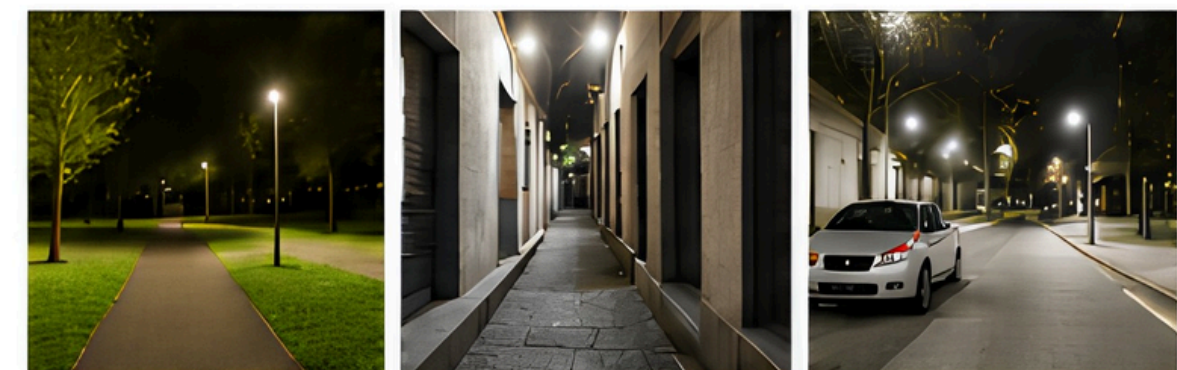


Sample 2

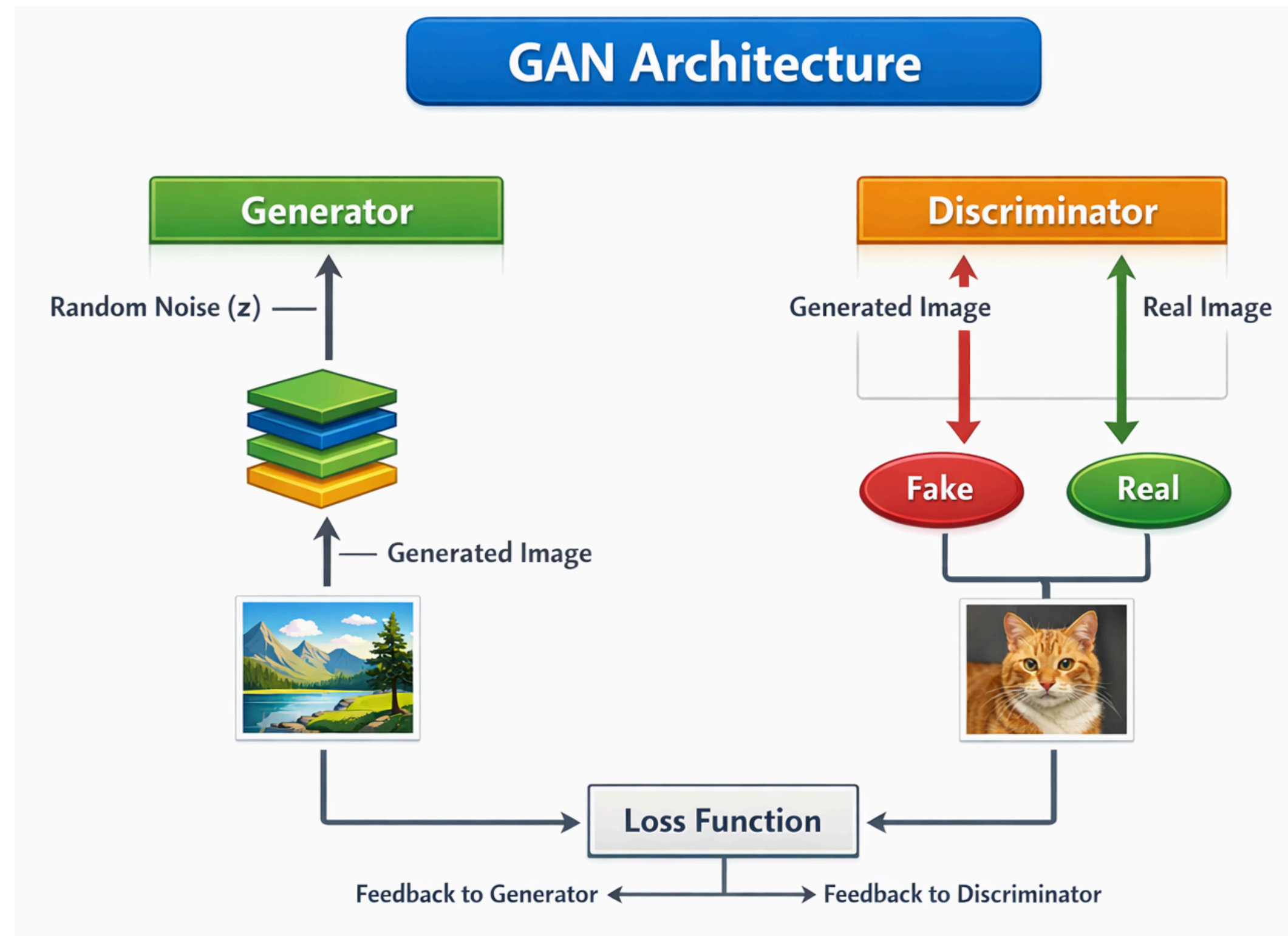
Low-light
Image



Ground-truth
Normal-light
Image



GAN Architecture & Loss Functions



Discriminator Loss

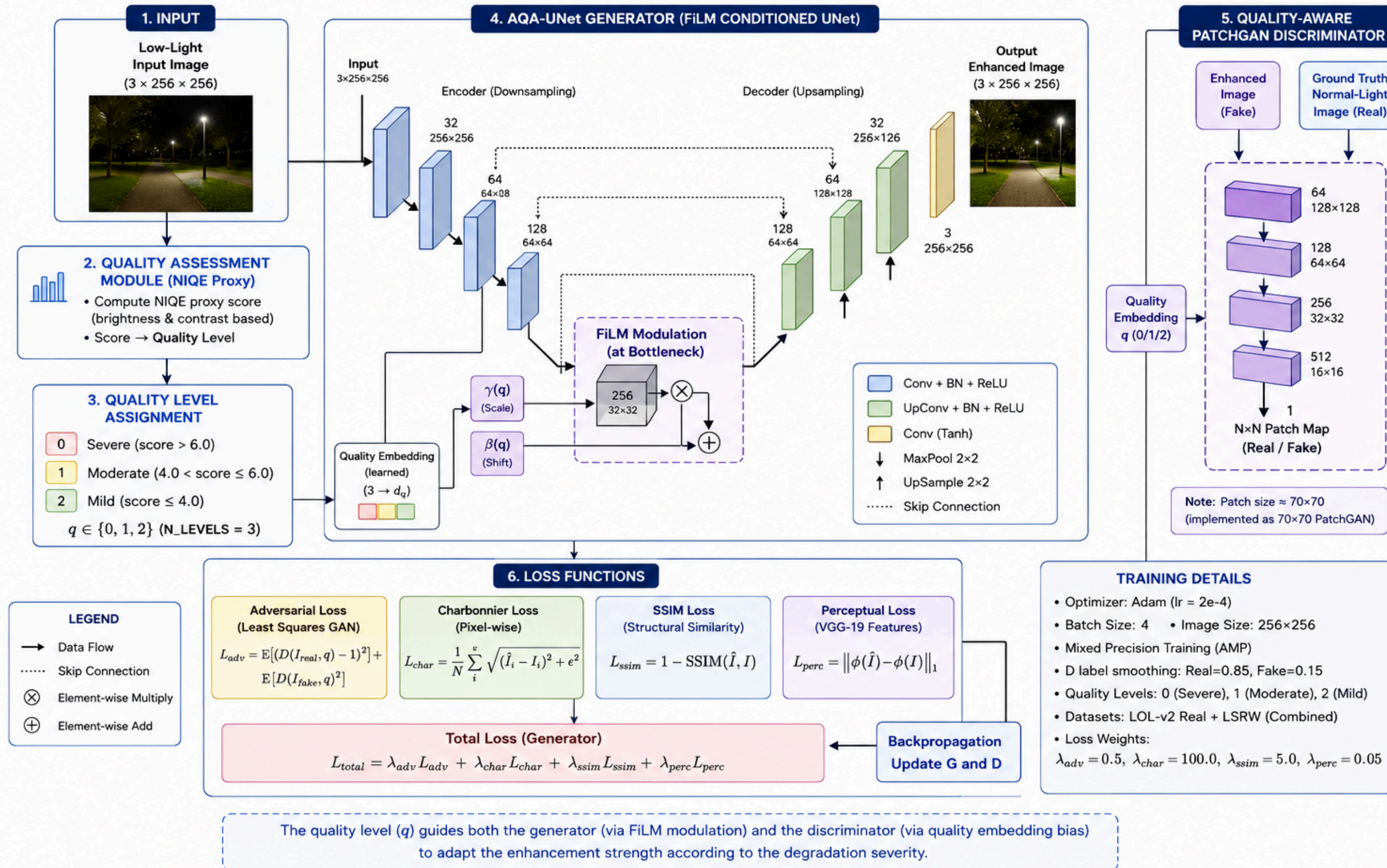
$$LD = -E_{x \sim p_{data}(x)}[\log D(x)] - E_{z \sim p_z(z)}[\log(1 - D(G(z)))]$$

Generator Loss

$$LG = -E_{z \sim p_z(z)}[\log D(G(z))]$$

Proposed Framework / Methodology

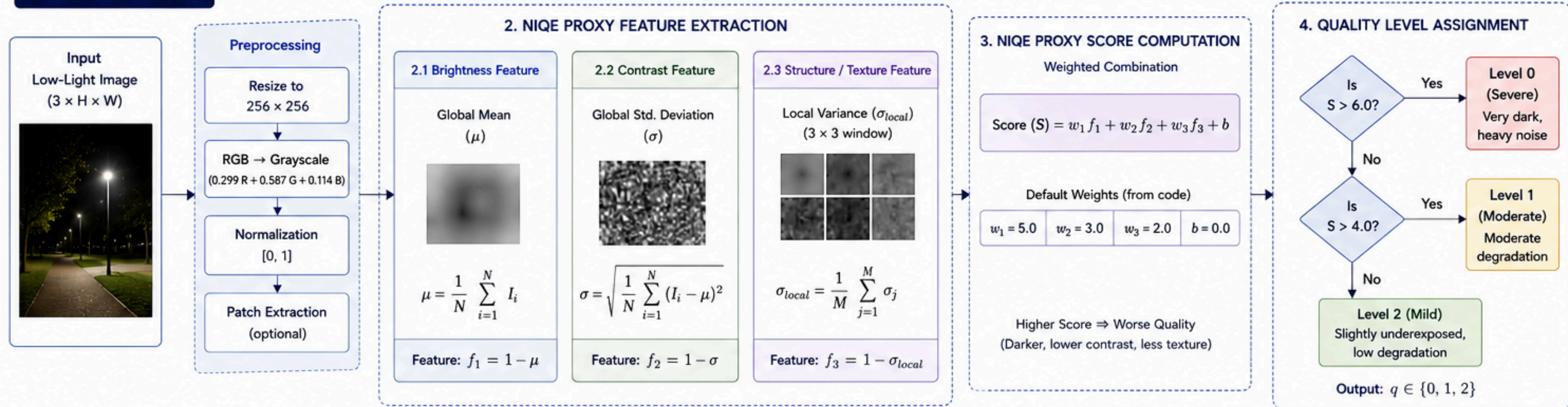
AQA-UNet: Adaptive Quality-Aware UNet GAN Architecture



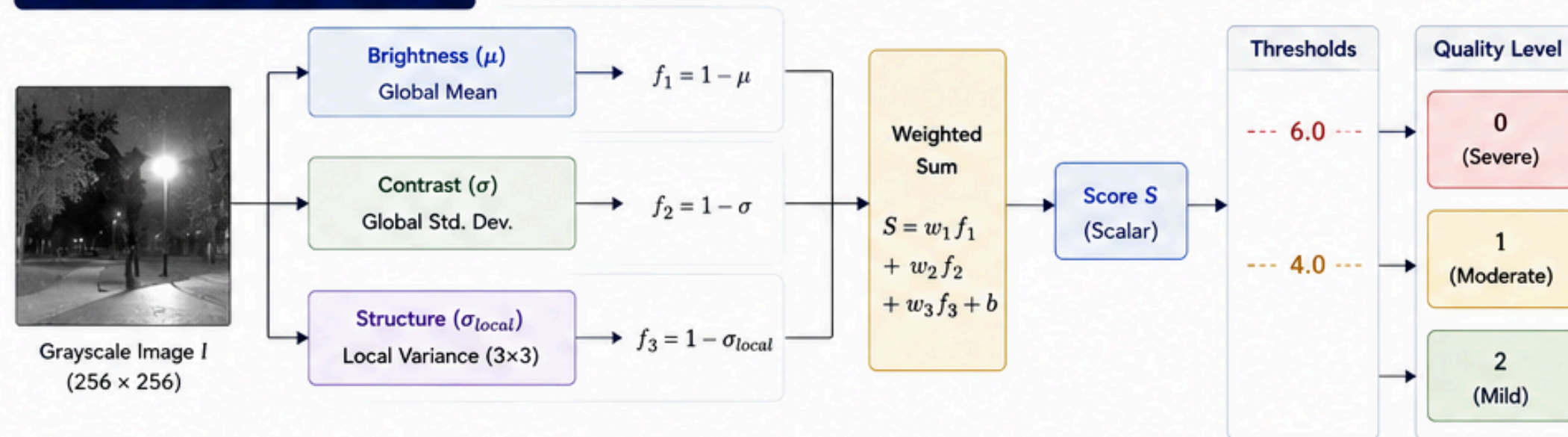
Proposed Framework / Methodology

QUALITY ASSESSMENT MODULE (QAM) – DETAILED ARCHITECTURE

1. OVERALL PIPELINE



2. DETAILED COMPUTATION FLOW

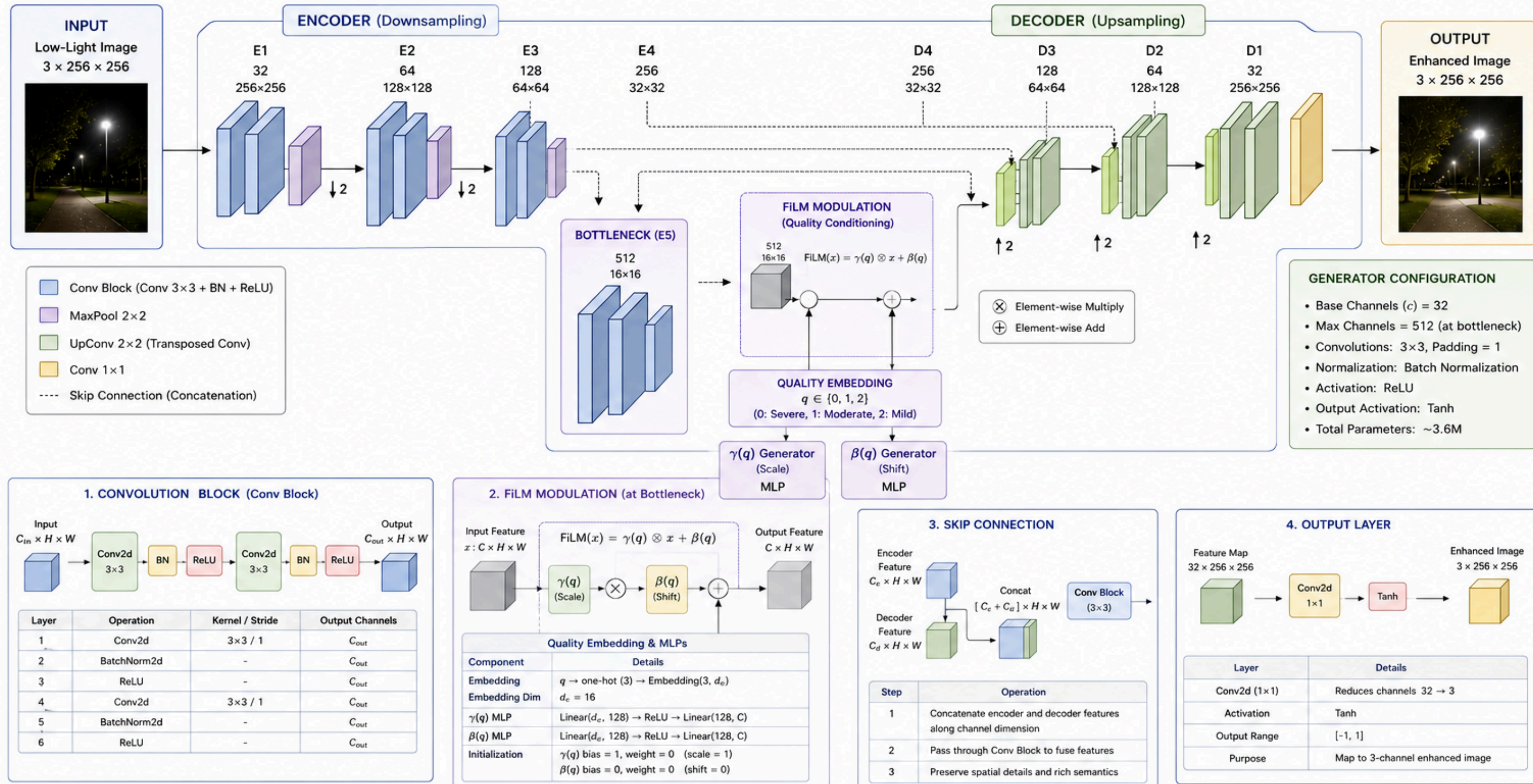


3. DESIGN CHOICES & CHARACTERISTICS

Component	Description / Rationale
No-Reference	QAM does not require any ground truth. Fast and training-free.
NIQE Proxy	Uses brightness, contrast and local structure statistics (approximation of NIQE behavior).
Lightweight	Very low computational cost (< 1 ms per image on GPU).
Robust	Works well across different scenes and illumination conditions.
Discrete Levels	3-level classification guides the generator and discriminator adaptively.
Differentiability	QAM is not used in backpropagation; it provides external conditioning (q).

Proposed Framework / Methodology

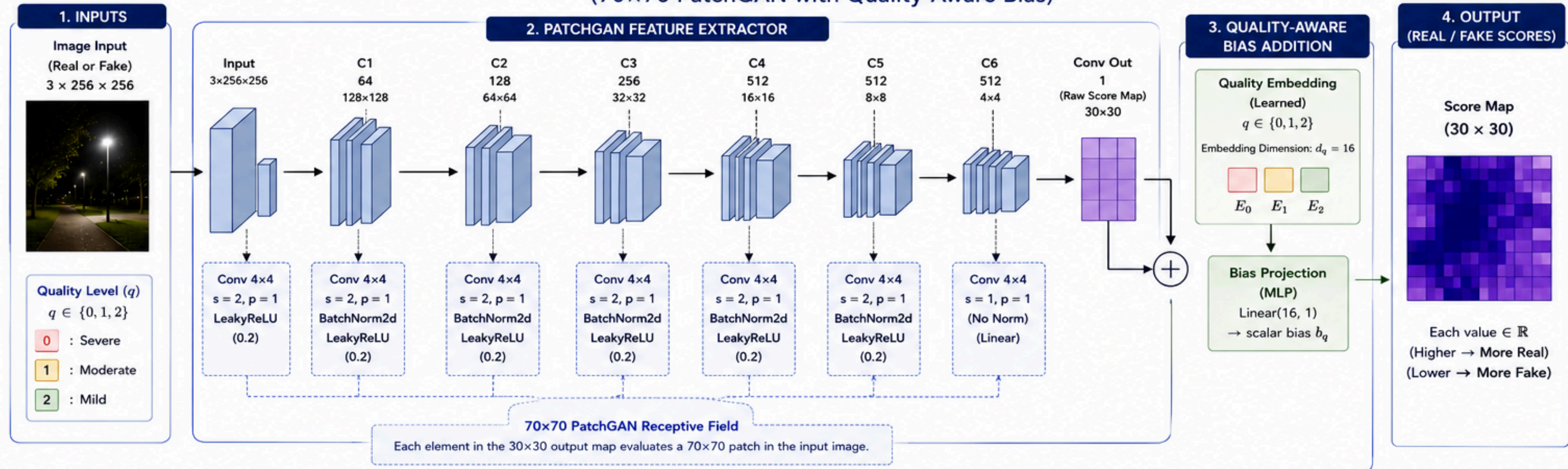
AQA-UNet GENERATOR – DETAILED ARCHITECTURE (FiLM Conditioned UNet)



Proposed Framework / Methodology

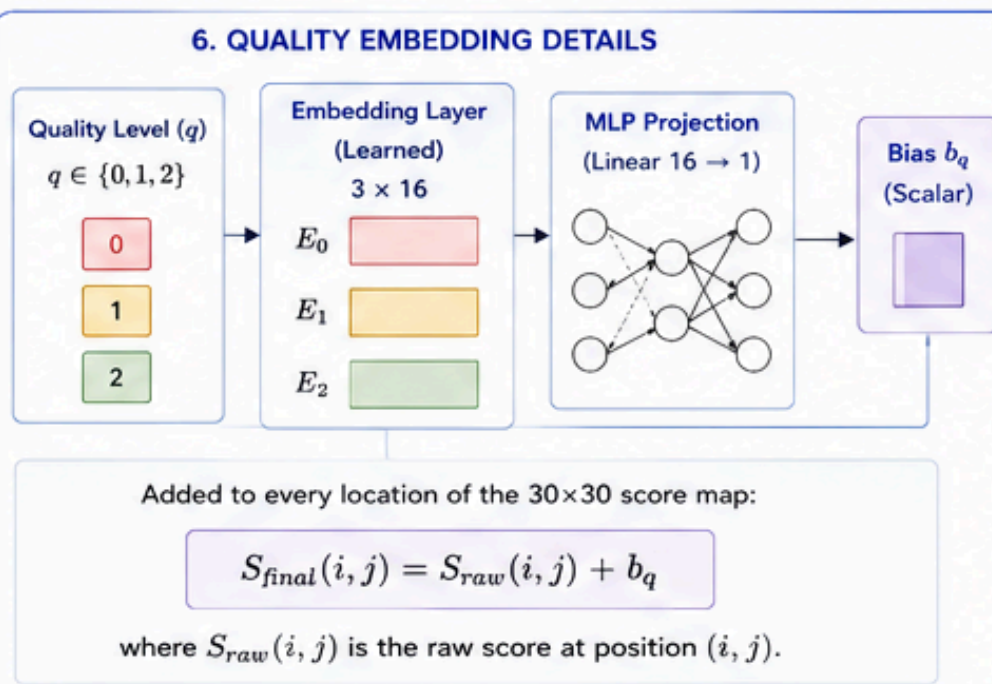
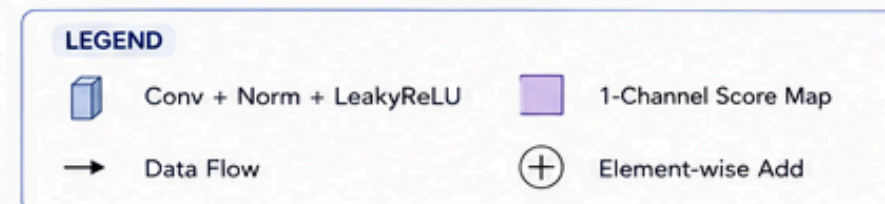
AQA-UNet DISCRIMINATOR – DETAILED ARCHITECTURE

(70×70 PatchGAN with Quality-Aware Bias)

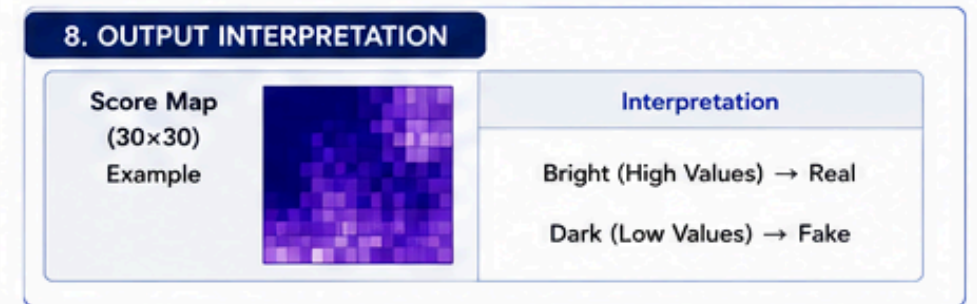


5. CONVOLUTIONAL BLOCK DETAILS

Layer	In Channels	Out Channels	Kernel	Stride	Padding	Norm	Activation	Output Size (H×W)
Input	3	64	4×4	2	1	–	LeakyReLU	128×128
C1	64	128	4×4	2	1	BN	LeakyReLU	64×64
C2	128	256	4×4	2	1	BN	LeakyReLU	32×32
C3	256	512	4×4	2	1	BN	LeakyReLU	16×16
C4	512	512	4×4	2	1	BN	LeakyReLU	8×8
C5	512	512	4×4	2	1	BN	LeakyReLU	4×4
C6 (Out)	512	1	4×4	1	1	–	Linear	30×30



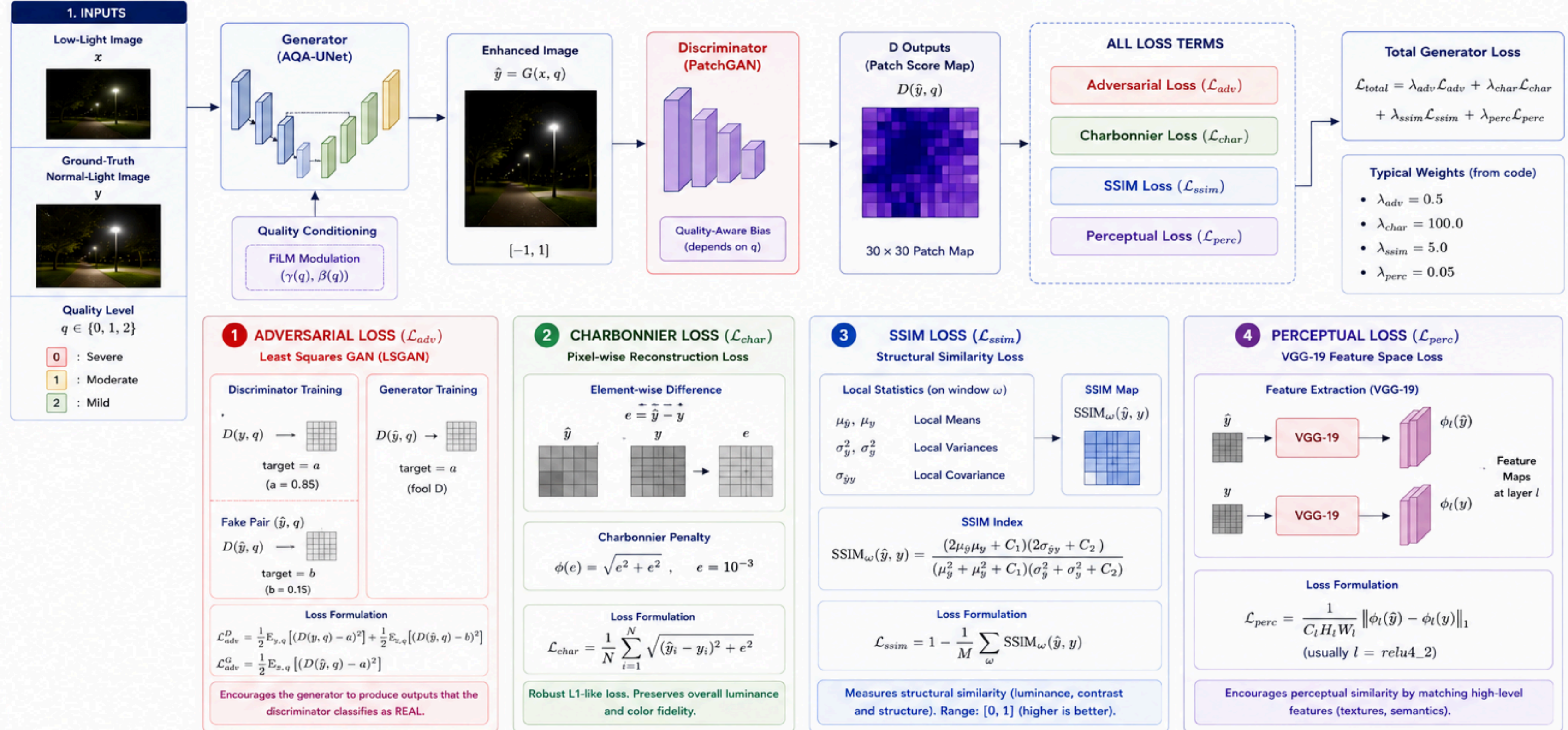
- 7. DESIGN CHARACTERISTICS**
- PatchGAN (70×70): Captures high-frequency local realism.
 - Spectral-friendly: Strided conv with LeakyReLU for stable training.
 - Quality-Aware: Adds learnable bias conditioned on degradation level.
 - Lightweight: ~ 2.7M parameters (base_ch = 64).
 - Outputs a real/fake probability map instead of a single scalar.



Proposed Framework / Methodology

AQA-UNet: LOSS FUNCTIONS ARCHITECTURE

Multi-Objective Loss for Quality-Aware Low-Light Image Enhancement



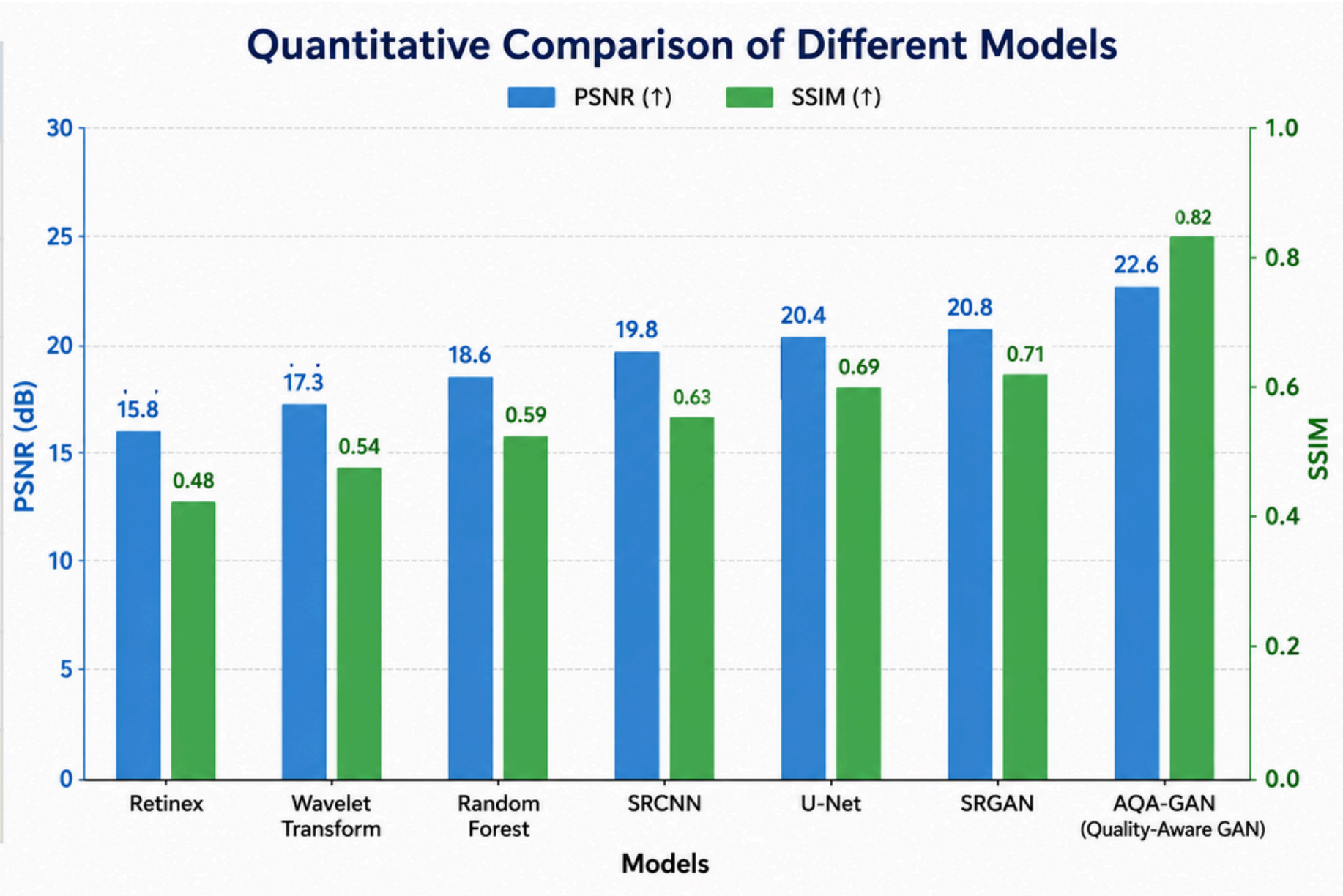
Experimental Results

Observation

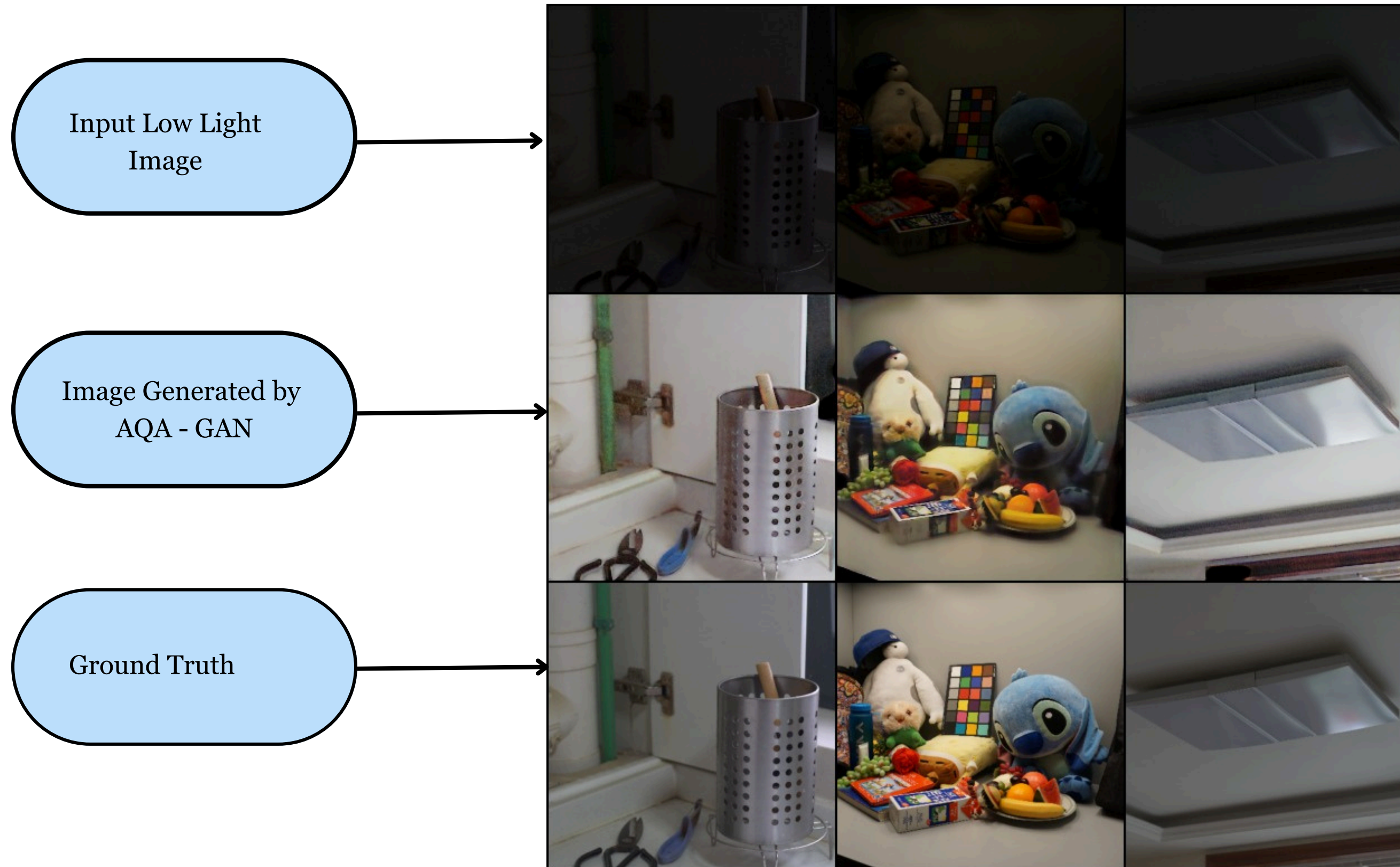
AQA-GAN achieves:

- Highest PSNR
- Highest SSIM
- Better texture preservation
- More natural enhancement

Model	Type	PSNR ↑	SSIM ↑
Retinex	Classical	15.8	0.48
Wavelet	Transform	17.3	0.54
Random Forest	Shallow ML	18.6	0.59
SRCNN	CNN	19.8	0.63
U-Net	CNN	20.4	0.69
SRGAN	GAN	20.8	0.71
AQA-GAN	Quality-Aware GAN	22.6	0.82



Experimental Results



Conclusion

AQA-GAN introduces adaptive quality-aware enhancement for nighttime images.

The proposed system:

- Dynamically adjusts enhancement intensity
- Produces visually realistic outputs
- Improves PSNR and SSIM significantly
- Preserves natural textures and colors

The framework successfully overcomes limitations of traditional uniform enhancement methods.

Future Direction



1. Continuous Quality Conditioning

- Replace discrete quality levels with continuous adaptive conditioning.
- Enable finer enhancement control for varying illumination conditions.

2. Real-Time Deployment on Edge Devices

- Optimize the model for real-time inference.
- Deploy on resource-constrained platforms.
- Reduce computational complexity and memory footprint.

3. Video Low-Light Enhancement

- Extend AQA-GAN from images to video sequences.
- Maintain temporal consistency and stable brightness.
- Reduce flickering and improve visual continuity.

5. Multi-Task Learning

- Develop a unified framework that jointly learns multiple enhancement tasks for better generalization and efficiency.

Multi-Task Learning: Key Tasks

Denoising

- Remove sensor noise and illumination noise.
- Improve image clarity in extreme low-light conditions.

Super-Resolution

- Recover high-frequency details.
- Enhance spatial resolution for sharper outputs.

Deblurring

- Restore sharp edges and textures.
- Correct motion blur and focus issues.

Thank You



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Retinex · Wavelet Denoising · Random Forests

1 Retinex

Models human vision: $\text{image} = \text{Illumination} \times \text{Reflectance}$. Multi-Scale Retinex (MSR) averages Gaussian blurs at $\sigma=15, 80, 250$ in log domain to estimate and remove illumination.

$$\log(R) = \log(I) - \log(L) \text{ where } L = \text{Gaussian}(I)$$

2 Wavelet Denoising

Decomposes image into frequency subbands. Noise concentrates in HH detail coefficients. BayesShrink adaptive soft-thresholding removes noise while preserving edges.

$$\tau = \sigma^2_{\text{noise}} / \sigma_{\text{signal}} \rightarrow \text{soft_thresh}(\text{coeff}, \tau) = \text{sign}(c) \cdot \max(0, |c| - \tau)$$

3 Random Forests

Trains an ensemble of 50+ decision trees on paired LR–HR image patches (5×5 features). Each tree specializes on a random subset; predictions are averaged.

$$\hat{y} = (1/T) \sum_i f_i(x) \text{ where } f_i = \text{decision tree } i, T = \text{total trees}$$

SRCNN • U-Net

4 SRCNN

First end-to-end CNN. Three convolutional layers replace the full sparse-coding pipeline: patch extraction (9×9), non-linear mapping (1×1), reconstruction (5×5). Trained with MSE loss.

$$f_1: \text{Conv}(9 \times 9, 64) \rightarrow f_2: \text{Conv}(1 \times 1, 32) \rightarrow f_3: \text{Conv}(5 \times 5, 1) \quad \text{Loss: } \text{MSE}(\hat{I}, I_{\text{gt}})$$

5 U-Net

Encoder-decoder with skip connections. Encoder progressively halves spatial resolution ($\times 4$) while doubling channels. Skip connections copy encoder feature maps to decoder, preserving spatial detail.

$$d_i = \text{ReLU}(\text{BN}(\text{ConvT}(d_{i-1}) \oplus e_{n-i})) \quad \text{where } \oplus = \text{channel-wise concatenation}$$

What is a Generative Adversarial Network(GAN)?

Core Concept (Goodfellow et al., 2014)

A GAN trains two neural networks simultaneously in a minimax game:

- **Generator G** — takes random noise z and produces synthetic data $\hat{G}(z)$ that tries to fool the Discriminator

- **Discriminator D** — classifies inputs as real or fake, providing feedback to improve G

They compete: G gets better at generating, D gets better at detecting, until G produces outputs indistinguishable from real data.

Minimax Objective:

$$\min_G \max_D V(D,G) = E[\log D(x)] + E[\log(1 - D(G(z)))]$$

Why GANs Are Hard to Train

Mode Collapse

The generator produces a limited variety of outputs and ignores most of the real data distribution — finding a few "safe" samples that fool the discriminator.

Training Instability

G and D must remain balanced at all times — if D becomes too strong, G receives no useful gradients and training collapses.

Vanishing Gradients

Early in training, D easily rejects fakes, causing $\log(1 - D(G(z))) \approx 0$ — the generator receives nearly zero gradient signal and learns nothing.

SRGAN - Super Resolution GAN

SRGAN (Ledig et al., 2017)

What It Is

SRGAN was the first GAN applied to image super-resolution. It combines **adversarial training** with a **perceptual VGG loss** to produce photo-realistic outputs — addressing the blurring problem of all MSE-based methods.

Generator Architecture

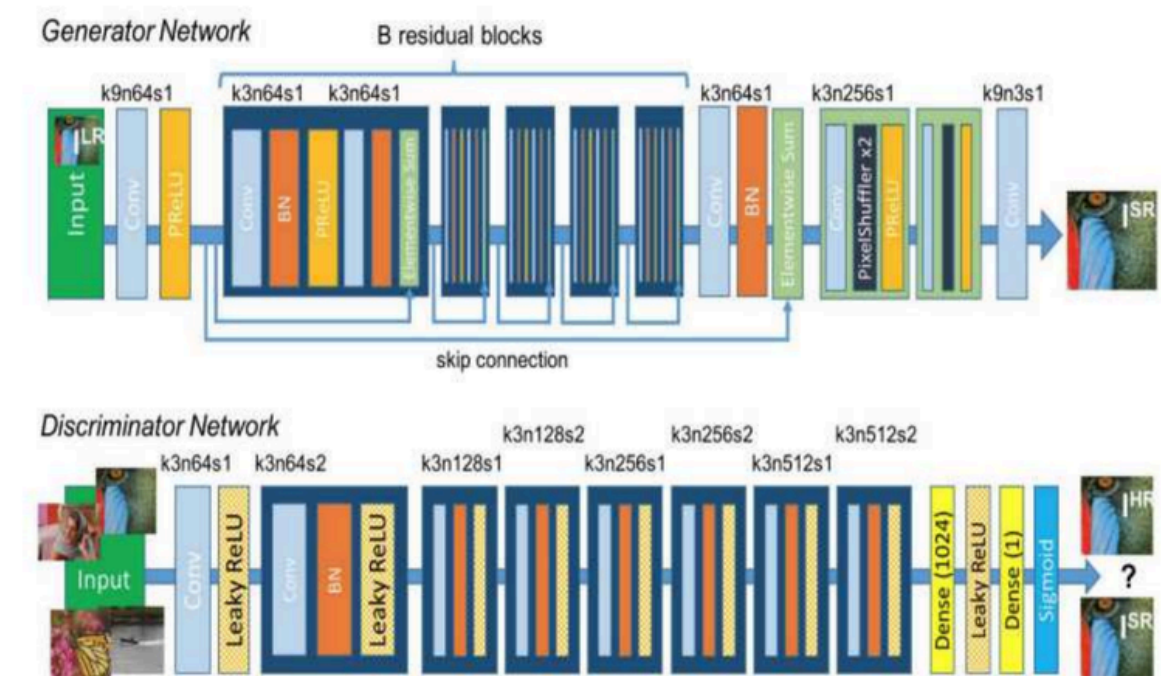
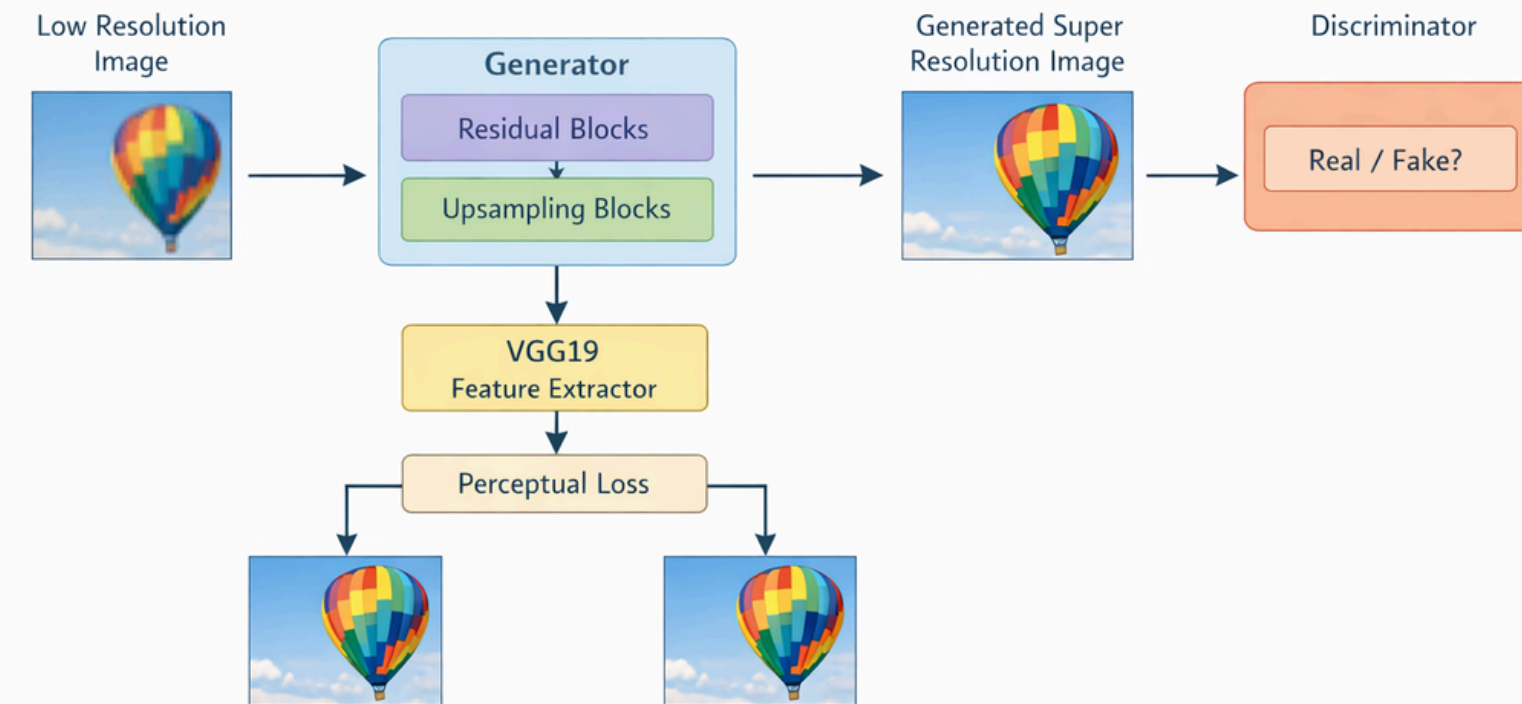
- 16 Residual Blocks — each with Conv→BN→PReLU→Conv→BN + skip connection
- Sub-pixel convolution (PixelShuffle) for 4× upsampling
- PReLU (Parametric ReLU) activation (learned negative slope)
- No BatchNorm in final layer — prevents artifacts

Total Loss Function

$$L_{total} = L_{content} + 10^{-3} \cdot L_{adv}$$

$$L_{content} = \| \psi(\hat{I}) - \psi(I_{gt}) \|_2^2 \text{ (VGG feature map MSE)}$$

SRGAN Architecture



Results and Discussions



Model	Type	Year	PSNR ↑	SSIM ↑	NIQE ↓	BRISQUE ↓	Key Strength
Retinex	Classical	1977+	15.8	0.48	5.92	43.1	No training needed
Wavelet	Transform	1990s	17.3	0.54	5.41	39.8	Preserves edges
Rnd Forest	Shallow ML	2012	18.6	0.59	4.97	36.2	First learned method
SRCNN	CNN	2014	19.8	0.63	4.52	33.5	End-to-end learning
U-Net	CNN	2015	21.4	0.70	4.18	31.8	Best spatial detail
SRGAN	GAN	2017	22.4	0.74	3.88	31.4	Photo-realistic textures

Classical Era Takeaway

Retinex and Wavelet require no training but cannot generalize — artifacts and fixed parameters limit performance on diverse ExDark scenes.

Deep Learning Gap

U-Net achieves the best pixel-fidelity among non-GAN models but produces blurry textures due to MSE loss. SRCNN and DnCNN face the same limitation.

GAN Advantage

SRGAN surpasses all non-GAN models on every metric. Adversarial + perceptual loss combination produces **sharper, more natural-looking** enhanced images.

Qualitative Results



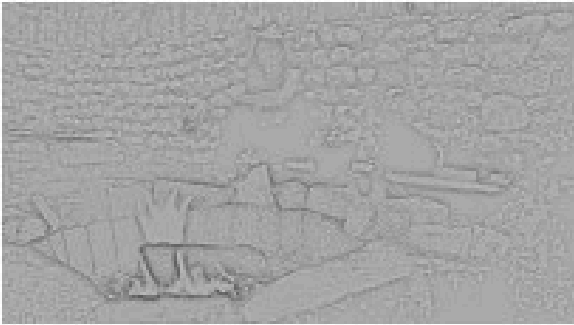
Original Dark Image



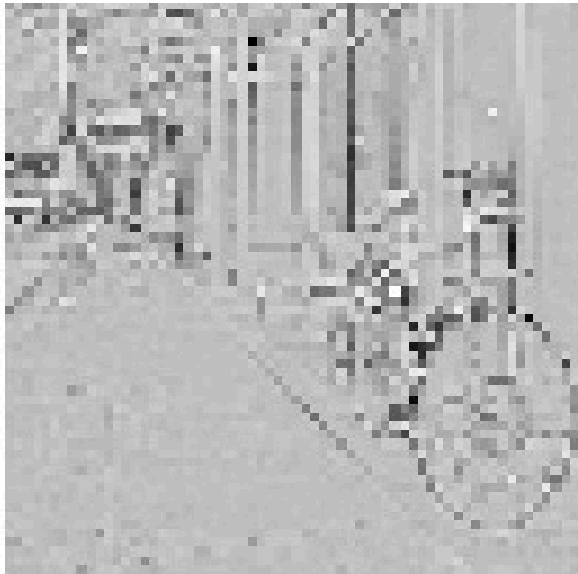
Retinex



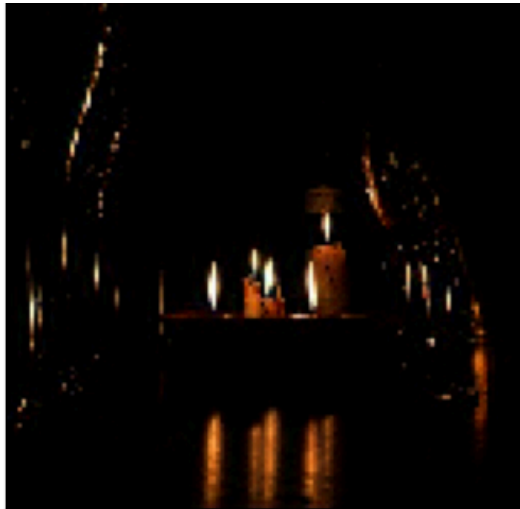
Wavelet



RF Prediction: SPIC



SRCNN



PSNR: 34.98 dB | SSIM: 0.9325



U Net



PSNR: 33.46 dB | SSIM: 0.9395



Ground Truth
256x256



SRGAN
256x256



Input
64x64



SRGAN